

ITT320 INTRO TO COMPUTER SECURITY

Chapter 3 : Fundamental of Firewalls

Zulazeze Sahri, UiTM



Objectives

- Explain how firewalls work
- Evaluate firewall solutions
- Differentiate between packet filtering and stateful packet filtering
- Differentiate between application gateway and circuit gateway
- Understand host-based firewalls and router based firewalls

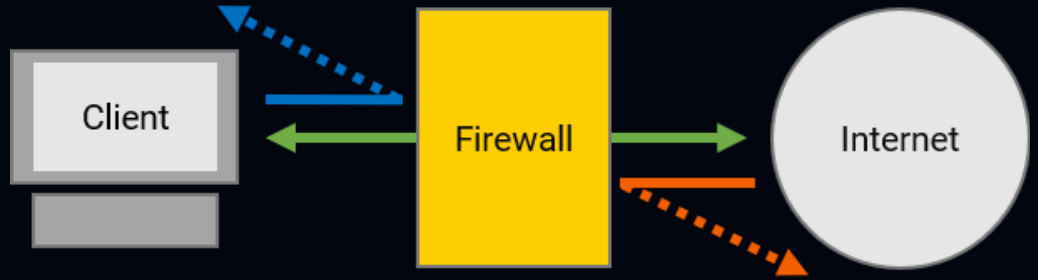


Introduction to Firewall

Firewalls are one of the fundamental devices used to secure a network.

This chapter explores the basics of firewalls and how they work.

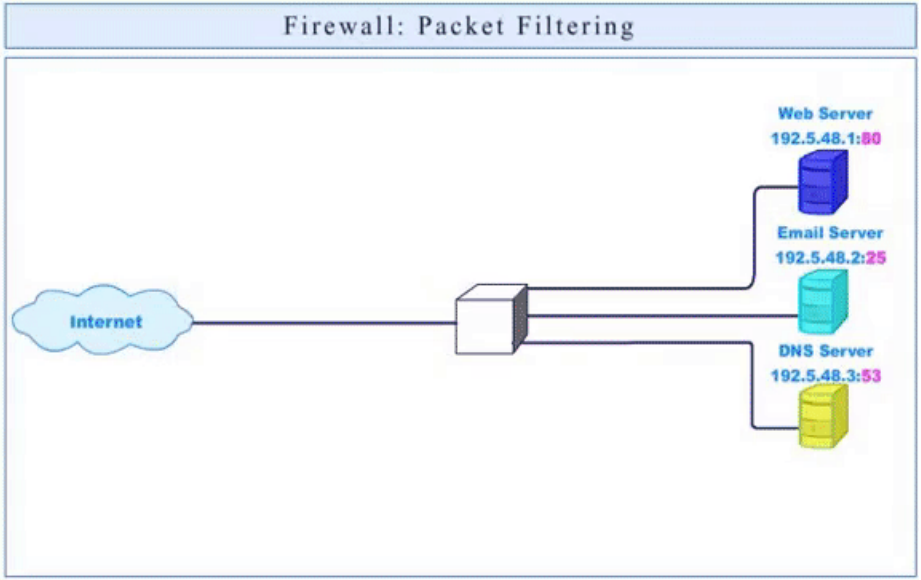
Based on the evaluations done here you will be able to determine under what circumstances a firewall solution is appropriate.



What is Firewall?

Firewall is a barrier between the world and your network

A **firewall** is a network security device that monitors incoming and outgoing network traffic and permits or blocks data packets based on a set of security rules.

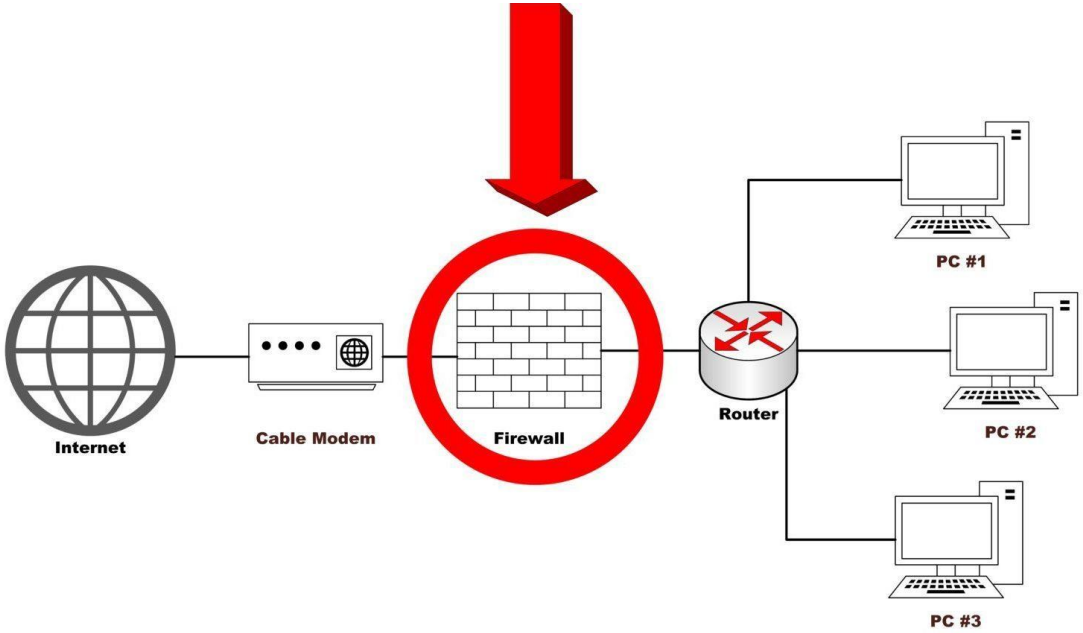


Can provide this barrier via:

- Packet filtering
- Stateful packet filtering
- User authentication
- Client application authentication

Where is Firewall Located?

In Between subnet , or before router, or else?





Types of Firewall

Major types of Firewalls

- Packet Filter
- Application Gateway
- Circuit Level Gateway
- Stateful Packet Inspection
- And More..

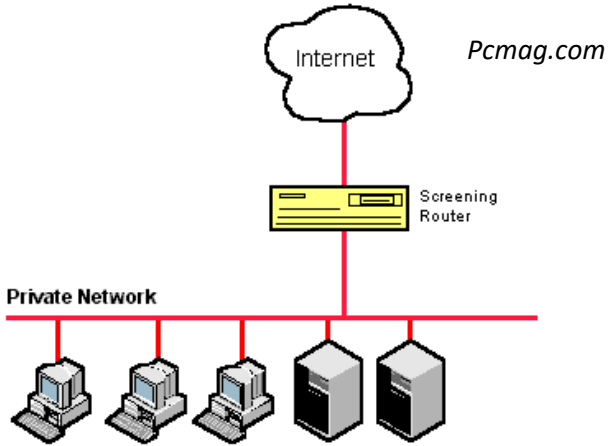
Types of Firewall
Packet Filter

Most basic type of firewall, also known as ‘Screening’ firewall

Operate on transport and network layer of TCP/IP

Works by examining a packet’s:

1. Source address
2. Destination address
3. Source port
4. Destination port
5. Protocol type

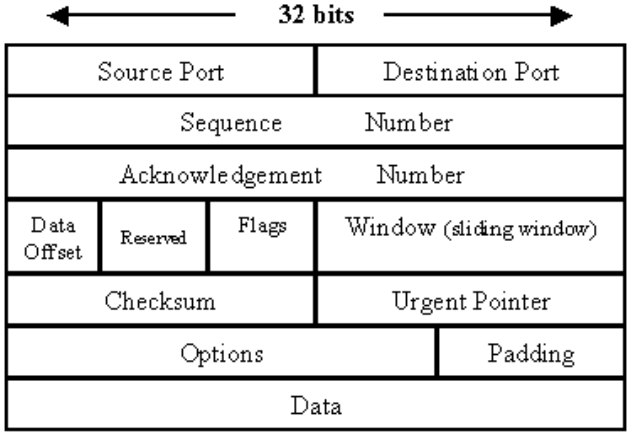


Example : Kapersky persolan AV/firewall, McAfee Personal AV/firewall, In Unifi Router

Types of Firewall
Packet Filter

Packet Filter Firewall Disadvantages:

- Does not compare packets
- No authentication
- Susceptible to SYN and Ping flood attacks
- Does not track packets
- Does not look at the packet data just the header
- Not necessarily the most secure firewall





Types of Firewall
Packet Filter

Packet Filter Firewall

We need to set the following firewall rules:

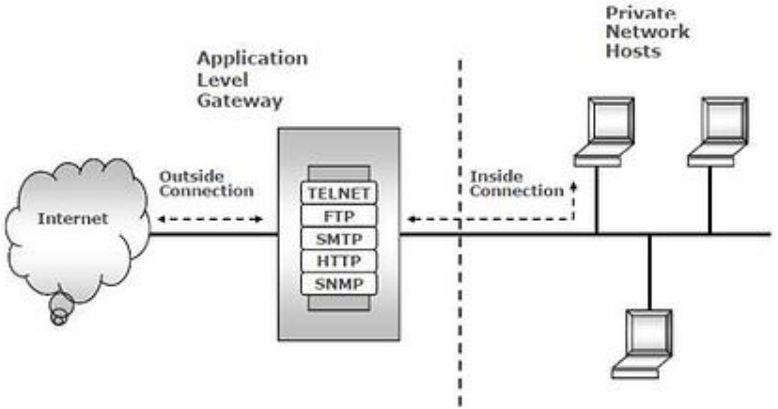
1. What types of protocols to allow (FTP, SMTP,POP3)
2. What source ports to allow
3. What destination ports to allow
4. What source IP addresses to allow

Types of Firewall
Application Gateway

Application Gateway

Also known as Application proxy or application-l
proxy. Operate on Application layer.

- Examines the connection between the client
  the server applications *(email,games,browser,messenger)*
- Enables administrators to specify what applications are allowed
- Allows for user authentication





Types of Firewall Application Gateway

The disadvantages of Application Gateway:

- Requires more system resources
- Susceptible to flooding attacks (SYN, Ping)
- Due to time it takes to authenticate user
- Once connection is made, packets are not checked

Example : MS Azure Applicate Gateway, Firebox by Watch box Technologies

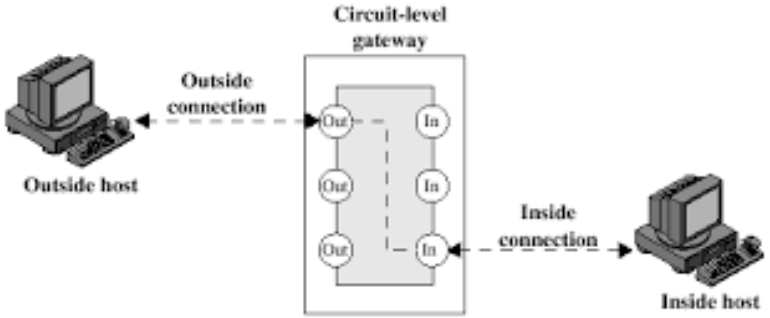
Types of Firewall
Circuit Level Gateway

Circuit-level gateways monitor TCP handshakes and other network protocol session initiation messages across the network

- More secure than application gateways
- Typically implemented on high-end equipment

How it works?

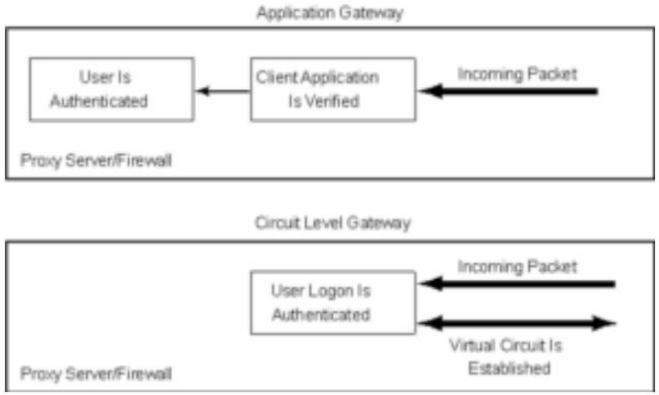
- Authenticates the user first
- Virtual circuit is used to pass bytes between client and proxy server
- External users only see the proxy IP not the internal client IP address



Types of Firewall
Circuit Level Gateway

Circuit Level Gateway

- External systems do not see internal systems
- May not work for some implementations
- Amrita Labs provides circuit level gateway software





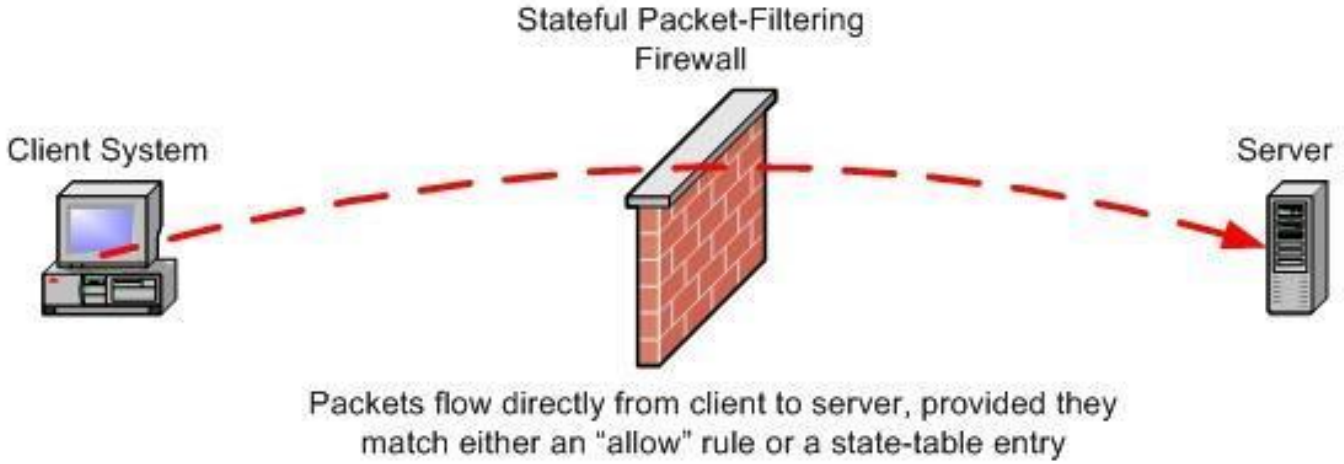
Types of Firewall
Stateful Packet Inspection

Monitors the state of active connections and uses this information to determine which network packets to allow through the **firewall**.

State-Aware Device : Not only examine packets

- Individually tracks sessions of network connection traversing
- Aware of context of packets makes them less susceptible to flood attacks
  - Knows if packet is part of a larger stream
  - Recognizes whether source IP is within the firewall
  - Can look at the contents of the packet

Types of Firewall
Stateful Packet Inspection





Types of Firewall
Stateful Packet Inspection

Stateful Packet Inspection

- ❑ Examples:
  - ✓ SonicWALL (www.sonicwall.com)
  - ✓ Linksys (www.linksys.com)
  - ✓ Cisco (www.cisco.com)

When possible, the recommended firewall solution



Hybrids Firewalls

Hybrid Firewalls

Becoming more popular, these configurations take multiple approaches to their firewall implementations

Example:

SPI and circuit level gateways might be used together



Implementing Firewalls

Need to understand the firewall's relationship to the network it is protecting

Most common solutions:

1. Network host-based
2. Dual-homed host
3. Router-based firewall
4. Screened host



Implementing Firewalls

Network Host-Based

A **firewall** installed on each individual server that controls incoming and outgoing **network** traffic

Software-based solution runs on top of operating system

Must harden the operating system in the following ways:

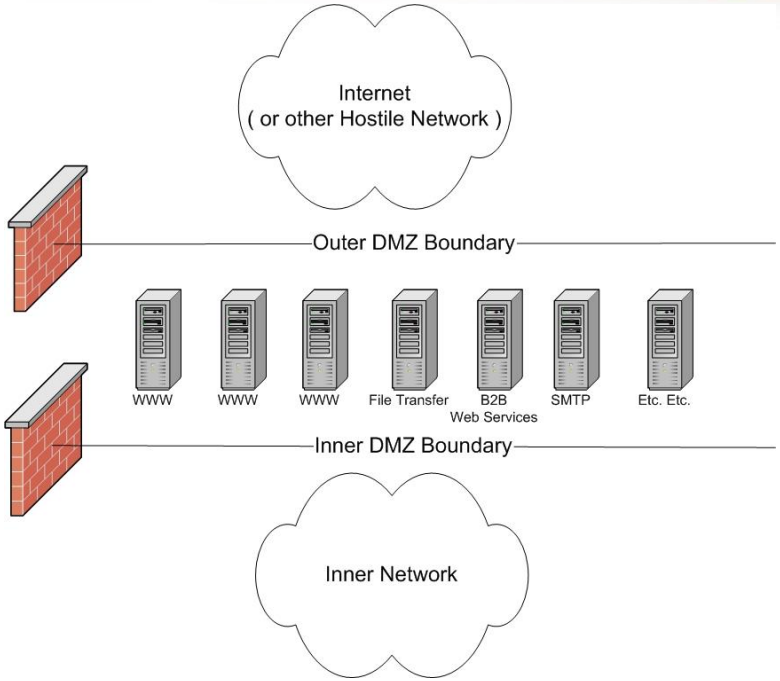
- Ensure all patches are updated
- Uninstall unneeded applications or utilities
- Close unused ports
- Turn off all unused services
  Cheap solution

Implementing Firewalls

In Practice - DMZ

Demilitarized zone

More companies are using these as part of their overall security solution

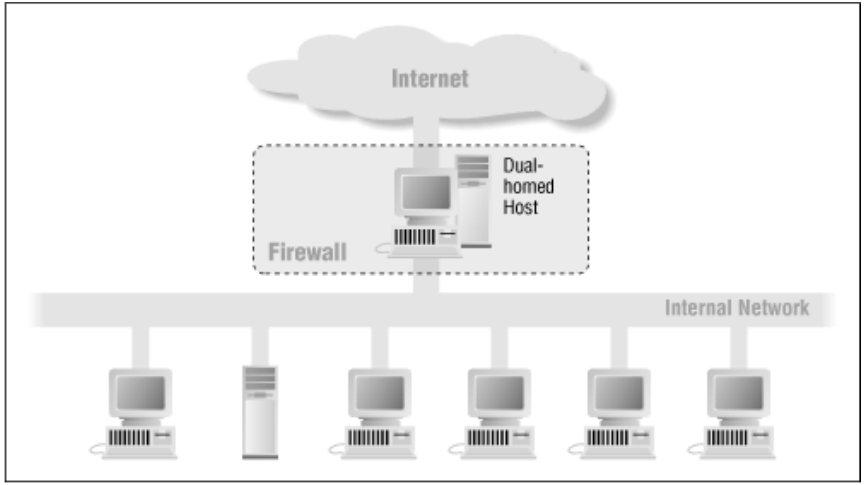


Implementing Firewalls

Dual-Homed Hosts

Networked device is built with two network interface (NICs).

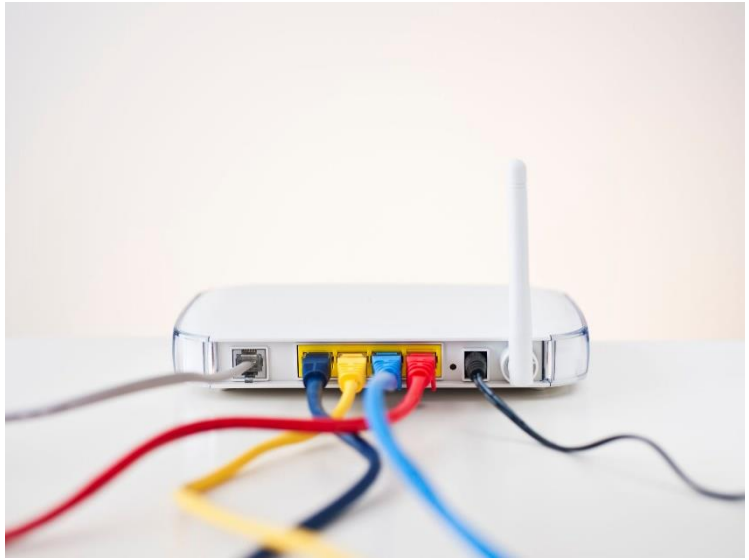
- Expanded version of the Network host firewall
- Also runs on top of an existing OS
- Disadvantage, like Network host firewalls, is its reliance on the security of the OS



Implementing Firewalls

Router Based Firewall

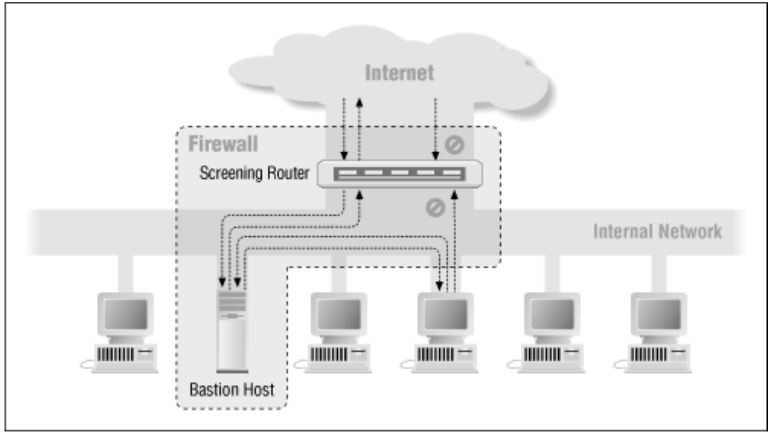
- Usually the first line of defense
- Uses simple packet filtering
- Ideal for novice administrators
- Can be preconfigured by vendor for specific needs of user
- Can be placed between segments of a network



Implementing Firewalls

Screened Host

- A combination of firewalls
- Bastion host and screening router is used
- Similar in concept to the dual-homed host

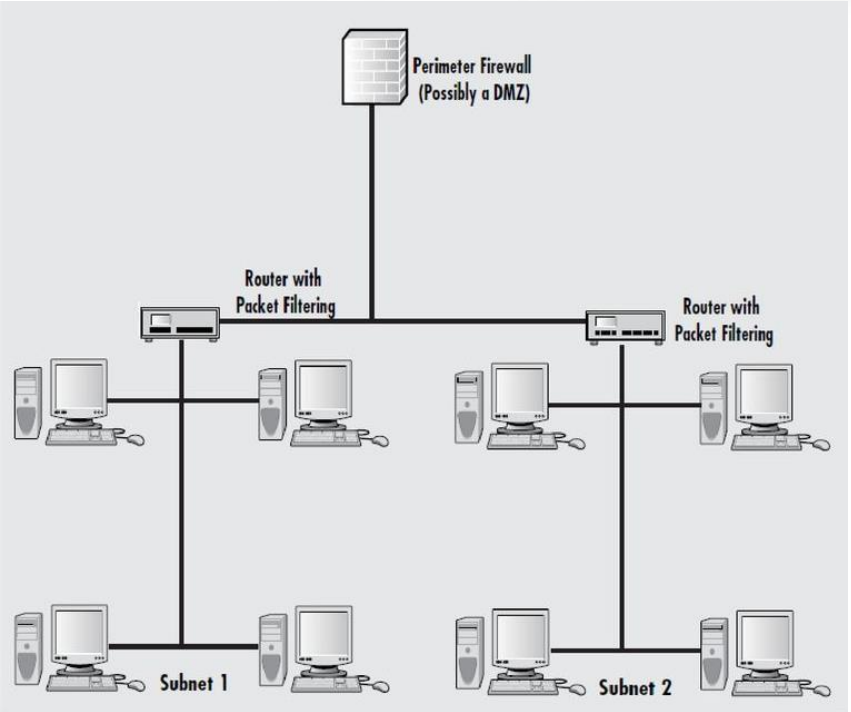


Implementing Firewalls

In Practice : Utmost Security

Organizations want the best security setup they
can get to ensure the protection of information

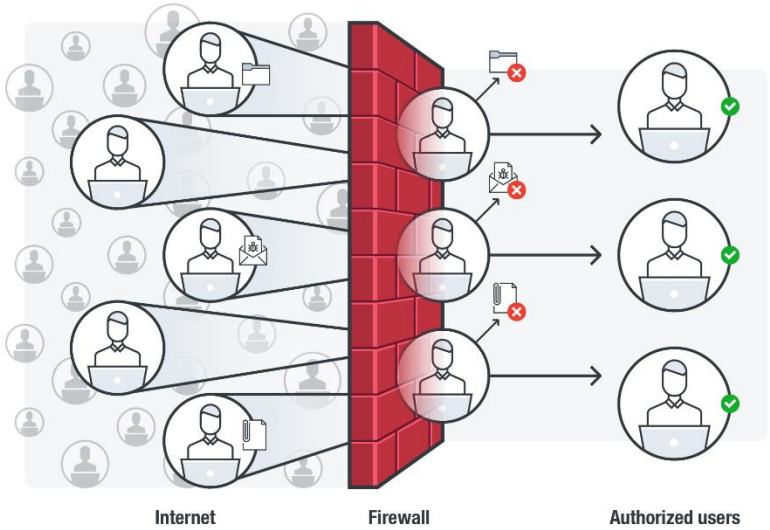
The graphic shows one setup beyond just a simple firewall



Selection & Using Firewalls

Using a Firewall:

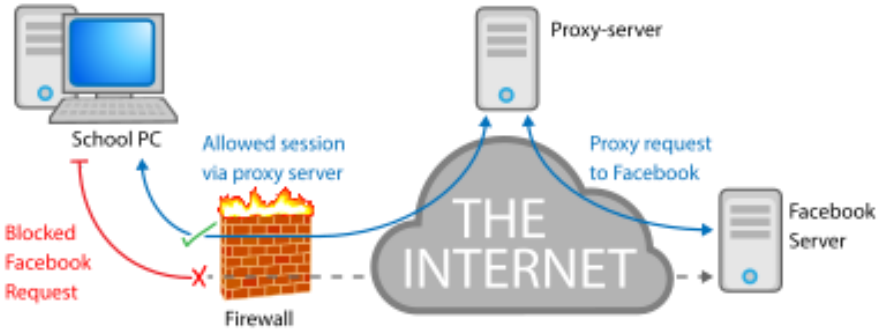
- Configure it properly
- Consider a consultant for initial setup
- Review logs periodically for anomalies
- Utilize statistics for baseline performance



Using Proxy Server

Prevent the outside world from gathering information about your internal network

- Provide valuable log information
- Can redirect certain traffic, based on configuration
- Typically runs on the firewall machine
- Protects against spoofing

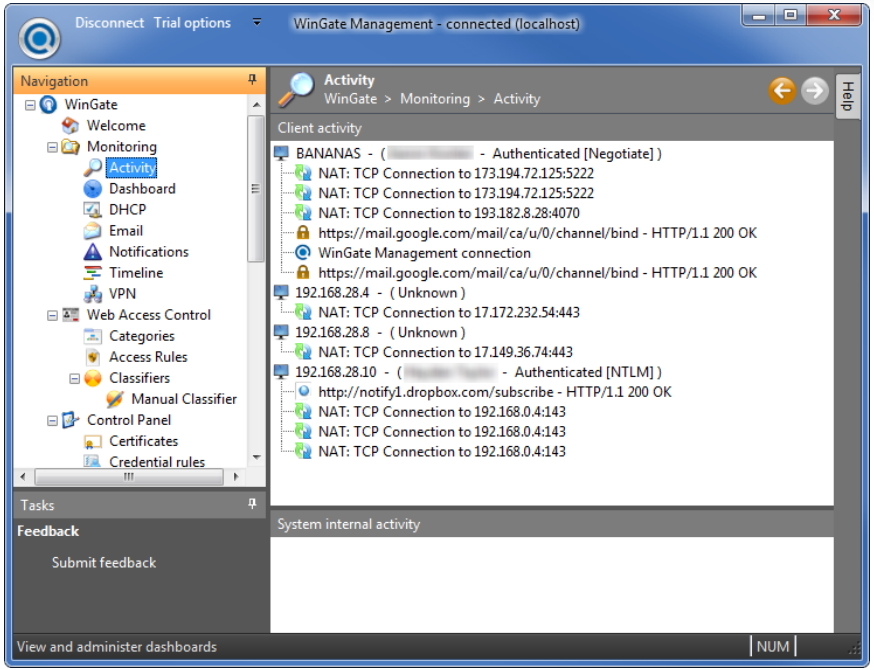


Using Proxy Server

The WinGate Proxy Server

Features include:

- Internet connection sharing
- Hiding internal IP addresses
- Allowing virus scanning
- Filtering of web sites
- https://www.wingate.com/products/wingate/index.php



Other brand ?



Using Proxy Server

Network Address Translation (NAT)

- Supersedes proxy servers
- Translates internal IP addresses to public addresses
- Can explicitly map ports to internal addresses for web servers



SUMMARY

- Firewalls and proxy servers are critical for network security solutions
- There are many solutions that can be considered
- Solutions range in price and features
- Should use most secure solution that budgets allow



SUMMARY

- Various types have been discussed:
  - Packet filter
  - Application gateway
  - Circuit level gateway
  - Stateful packet inspection

- Implementations include:
  - Network host-based
  - Router-based
  - Dual-homed and Screened host